Run report: GHS cover census n = 3 (larger q) and n = 5, run 20261014T0050Z

Descriptive record of one run. Interpretation is in the study `README.md` / `REPORT.pdf`. Conductor T-2467.

What ran

- Spec: `PROTOCOL.md`, frozen at commit `2d95bde3f` before any panel; amendment `338cd011a` (hypothesis H) after panel D and before panels A, B, C, E produced output. Protocol-versioned at commit, not independently preregistered.
- Tool: `cover_census_n.rs` (dependency-free Rust, `rustc --edition 2021 -O`), sha256 in the environment block. Orchestration `run.sh` (pilot, reproduction arms, panel D) and `run_panels.sh` (panels A, B, C, E); comparisons `compare.sh` (`jq` + `cmp`).
- Isolation: none. Shared Apple M4 Pro host, load average 60–260 throughout; 6 threads. Wall times are bookkeeping, not results. All recorded values are exact, deterministic counts.
- Seeds: fixed-seed stream of the merged tool (seed 20261012; tag q for n = 3, tag 100q + n for n = 5).
- Interruptions: the session that wrote the protocol ended during a timing pilot (`n = 5, q = 5, sample:1`, output unread); its scratch outputs, including the first `n = 3, q = 3` pilot records, were lost, and the pilot was rerun unchanged in this run. A later session interruption did not stop the running panel process. Panel E was killed once by a host restart about 2h15m into its first attempt, before it wrote any curve record; it was rerun from scratch with the same command and seed (recorded in the environment block). No other arm was restarted.
- Raw records: `results/run-20261014T0050Z/*.jsonl` (`rec` = field, curve, group, mr, derived_checks, totals), stderr files (all empty), `reproduction-checks.txt`.

```
run: 20261014T0050Z
commit (protocol frozen at): 18a7d6d5ff6ec8c7bb6b197ed9a2f0305310d4af
source sha256: d2a8504b68b26ea5ef83687d682a2bae75bb254acaf7d909420103e22aa11edc
protocol sha256: 2e9ed2a7b3c39583e518165e2fcf6dbd29611e16afa8105e09439a76a1c57f46
rustc: rustc 1.93.1 (01f6ddf75 2026-02-11) (Homebrew)
host: Darwin 25.6.0 arm64, Apple M4 Pro, 14 cpus
threads: 6
uptime at start: 17:47  up 12 mins, 4 users, load averages: 66.61 43.35 24.67
pilot-n3-q3-cosets: ccn 3 3 exhaustive cosets 6 -> exit 0, wall_s 0, load averages: 68.49 44.13 25.06
pilot-n3-q3-cocycles: ccn 3 3 exhaustive cocycles 6 -> exit 0, wall_s 0, load averages: 68.49 44.13 25.06
R1-n3-q5: ccn 5 3 exhaustive cosets 6 -> exit 0, wall_s 6, load averages: 69.01 44.64 25.35
R1-n3-q7: ccn 7 3 exhaustive cosets 6 -> exit 0, wall_s 42, load averages: 76.38 50.00 28.31
R2-n3-q11-sample120: ccn 11 3 sample:120 cosets 6 -> exit 0, wall_s 142, load averages: 92.03 66.50 38.31
D-n5-q3-cosets: ccn 3 5 exhaustive cosets 6 -> exit 0, wall_s 98, load averages: 61.23 62.89 39.90
R3-n5-q3-cocycles: ccn 3 5 exhaustive cocycles 6 -> exit 0, wall_s 3232, load averages: 164.91 158.34
155.35
reproduction arms done
panels started 20261014T0146Z from arm A; source sha256
d2a8504b68b26ea5ef83687d682a2bae75bb254acaf7d909420103e22aa11edc; commit
4189c22a857516797a5a60c30566ad94c0b54841; load averages: 164.62 158.78 155.62
A-n3-q11: ccn 11 3 exhaustive cosets 6 -> exit 0, wall_s 918, load averages: 151.84 141.91 144.63
B-n3-q13: ccn 13 3 exhaustive cosets 6 -> exit 0, wall_s 4956, load averages: 147.01 154.50 157.68
C-n3-q17-sample200: ccn 17 3 sample:200 cosets 6 -> exit 0, wall_s 1245, load averages: 155.82 148.85
147.20
INTERRUPTION: arm E-n5-q5-sample60 (started 20261014T0345Z) was killed when the agent session ended, about
2h15m in, with no curve record written (the tool emits records only at the end; only its field line
survived). No output of E was seen. E is restarted from scratch with the same command and fixed seed; the
partial file is overwritten. 20261014T0600Z
panels started 20261014T0600Z from arm E; source sha256
d2a8504b68b26ea5ef83687d682a2bae75bb254acaf7d909420103e22aa11edc; commit
e483c32555709199068d14c8fb39dd1315846ef4; load averages: 242.42 88.10 34.47
compile FAILED
note: the restart above failed to compile because the scratch bin directory no longer existed
(run_panels.sh does not create it); directory created, restart repeated. 20261014T0600Z
panels started 20261014T0601Z from arm E; source sha256
d2a8504b68b26ea5ef83687d682a2bae75bb254acaf7d909420103e22aa11edc; commit
e483c32555709199068d14c8fb39dd1315846ef4; load averages: 261.26 113.57 46.54
E-n5-q5-sample60: ccn 5 5 sample:60 cosets 6 -> exit 0, wall_s 4746, load averages: 121.29 144.86 151.83
uptime at end:  0:20  up  1:22, 32 users, load averages: 121.29 144.86 151.83
DONE
note (added after the run, 20261014T0730Z): host uptime at the E restart (up 25 mins at about 06:23Z) shows
the first E attempt was ended by a host restart, which also ended the agent session; the INTERRUPTION line
above names only the session. The commit line at the top is HEAD when the run started (18a7d6d, adds
run.sh); PROTOCOL.md was frozen at 2d95bde3f and not edited between that commit and 18a7d6d; amendment
338cd011a came after panel D.
```

Arms requested and completed

Arm	Requested	Completed	Structures per model	Notes
pilot	n=3, q=3, both methods	56 classes each	819	pilot, char 3
R1	n=3, q=5, 7, all classes	252 + 690	16275 / 120099	compared with merged run 20261013T0403Z
R2	n=3, q=11, sample 120	120 + 127 neighbours	1786323	compared with merged sample
D	n=5, q=3, all classes	488	597861	panel
R3	n=5, q=3, cocycle method	488	597861	compared with D
A	n=3, q=11, all classes	2662	1786323	panel
B	n=3, q=13, all classes	4400	4855539	panel
C	n=3, q=17, sample 200	200 + 187 neighbours	24221379	panel
E	n=5, q=5, sample 60	60 + 64 neighbours (after one restart: host restarted 2h15m into the first attempt, no records written; rerun from scratch, same seed)	254313125	panel

Reproduction checks

```
PASS R1 curve records q=5 (252 rows)
PASS R1 mod4 histograms q=5 (9 rows)
PASS R1 curve records q=7 (690 rows)
PASS R1 mod4 histograms q=7 (9 rows)
PASS R2 sampled curve records q=11 (120 rows)
PASS R3 cosets vs cocycles n=5 q=3 (488 rows)
PASS pilot cosets vs cocycles n=3 q=3 (56 rows)
```

Recorded values

Unit: number of isomorphism classes of curves (exhaustive arms) or sampled curves. `mismatches` (genus formula against the sum over quadratic subcovers) is 0 in every arm; `derived_checks` reports 0 violations in every arm. The plotted heights are recorded `group` counts divided by the recorded group size.

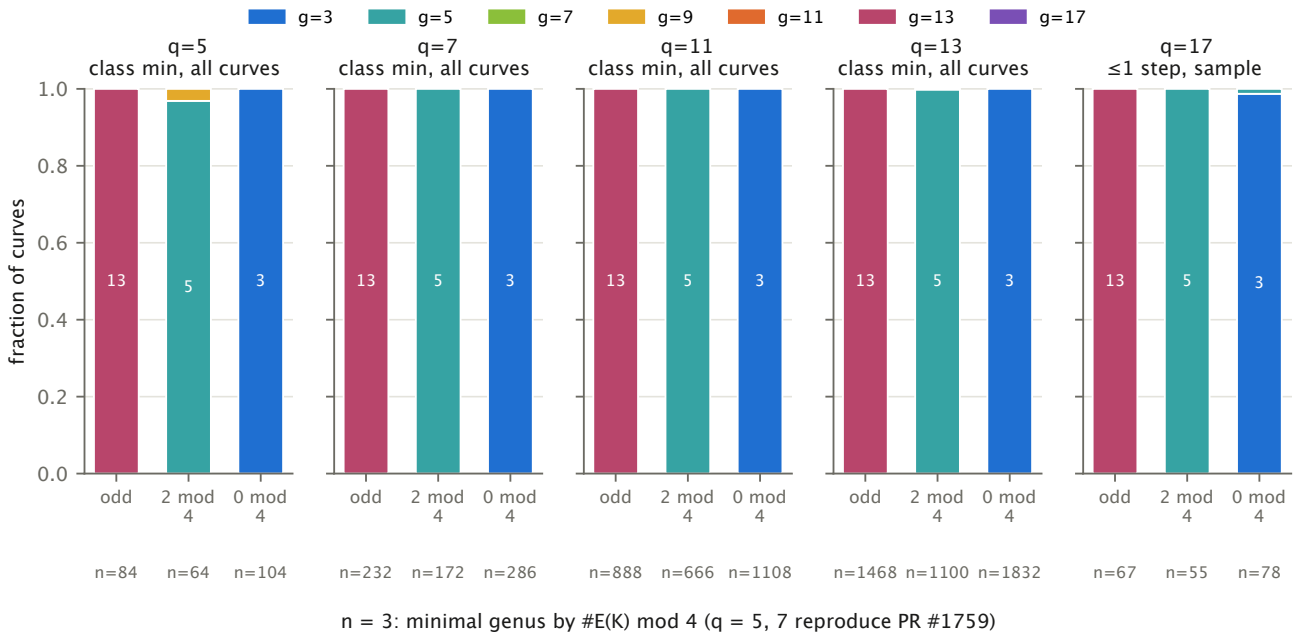


Figure 1: Run 20261014T0050Z, n = 3. Minimal genus over the isogeny class (q = 5, 7, 11, 13: every class) or within one 2-isogeny step (q = 17: sample of 200), by #E(K) mod 4; n under each bar is the recorded curve count.

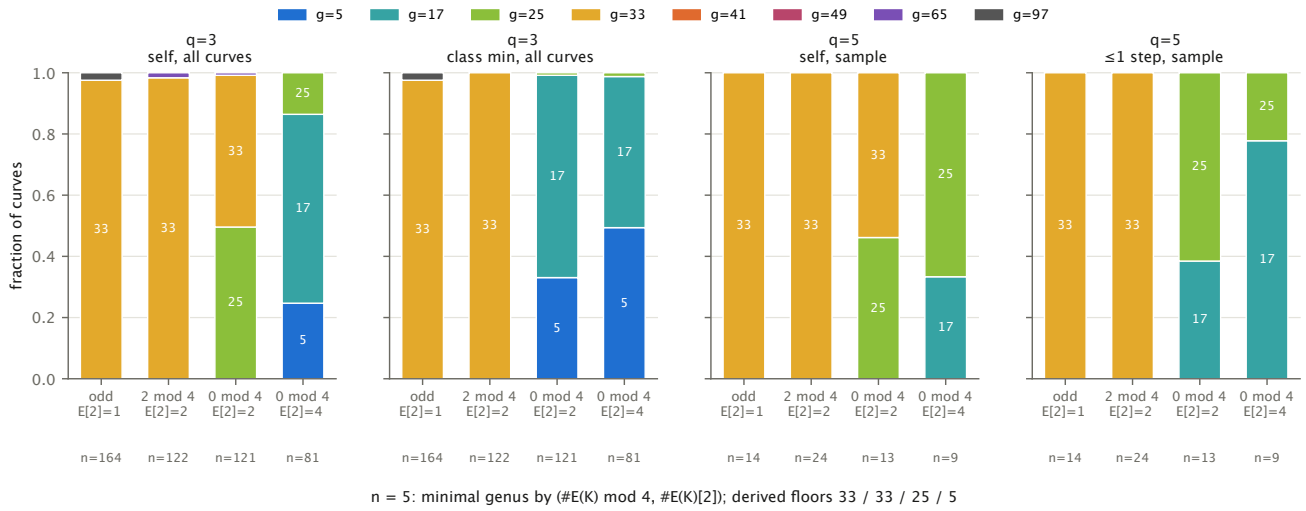


Figure 2: Run 20261014T0050Z, n = 5. Minimal genus by (#E(K) mod 4, #E(K)[2]); q = 3 every class, q = 5 sample of 60.

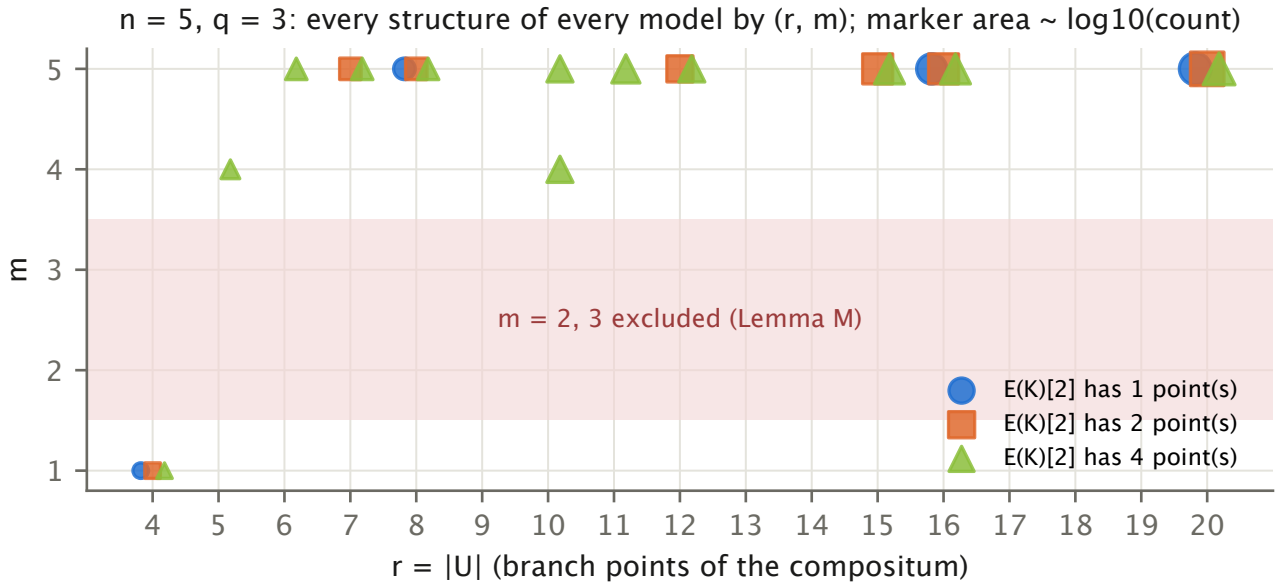


Figure 3: Run 20261014T0050Z, n = 5, q = 3: every structure of every curve by (r, m) and #E(K)[2], from the `mr` records.

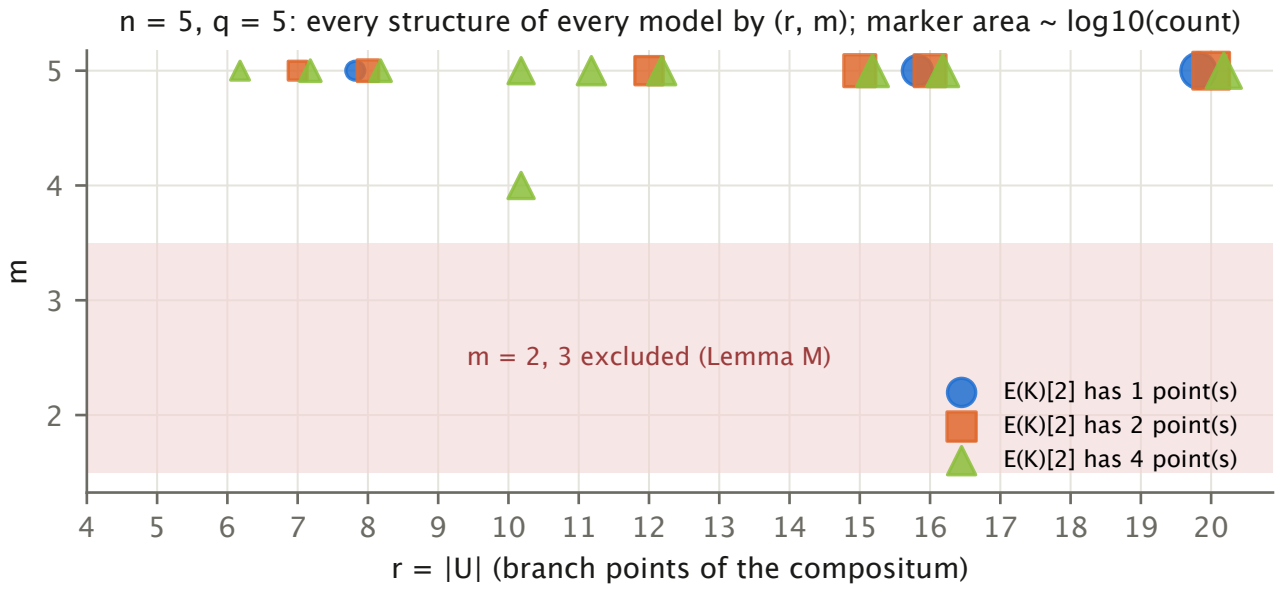


Figure 4: Run 20261014T0050Z, n = 5, q = 5: every structure of the 124 sampled and neighbour models by (r, m) and #E(K)[2].

Missing data

- Isogeny-class minima for the sampled arms C and E.
- Hyperellipticity of the n = 3, m = 2 genus-3 covers (not decided by branch data).
- No timing result of any kind.